Ainesh Chatterjee

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Education

University of Maryland - College Park

Dual BS in Computer Science (Machine Learning&Quantum Information), and Mathematics
May 2026 | GPA: 3.554
University, CS Departmental Honors; BS/MS; Dean's List

- **AI/ML**: Intro to: AI, Data Science, ML; Graduate NLP; Advanced Linear Algebra
- **Math**: Calc III; Linear Algebra; Differential Equations; Advanced Calculus; Abstract Algebra
  Mathematical Finance: Derivatives & Stochastic Models
- **CS**: Algorithms; Data Structures; Computer Systems; Object-Oriented Programming; Organization of Languages
- **Stat**: Applied Prob&Stat; Probability Theory

Projects

Vizier (active) | Team Lead/ML Developer

- AI-powered platform for personalized newsletters
  (Full Pipeline: Content Aggregation → Monetization)
- Test-Time MoE agentic architecture for improved context retrieval via specialized document-expert LLM models

QSafe | Solo Developer (active)

- Open-Source Python/Rust Quantum-Safe password manager with lattice-based cryptography
- Secure Docker container core manager
- End-to-end encrypted CLI-container comm protocol

CoronaSafe | Team Lead/Backend Developer

- Python/Flutter app for global COVID-19 risk assessment
- Analyzed real-time foot traffic and urban density using a time-weighted algorithm for predictive accuracy
- **Congressional App Challenge Winner**: 2021 District MD08
- **Guest Speaker** at 2022 US Patent and Trademark Office APPLY Yourself event

Resourceful | Team Lead/Backend Developer

- Python/Flutter app that connected underrepresented students to resources using NLP-driven searches
- Implemented advanced NLP techniques
  (e.g. NLTK, Spacy, and Cosine, Wu-Palmer similarity)
- **Best Education Award**: 2022 Blairhacks_5 Hackathon

Skills

- **Programming**: Python, C/C++, DevOps, Webhosting, Fullstack Development, API-creation, Design Paradigms
  - Familiar: Java, Rust, Lua, MATLAB, Flutter/Dart, HTML5, CSS3, JavaScript, Assembly
- **ML/AI**: Un/Supervised Learning, Deep RL, GANs
- **Data Science**: Statistical Analysis, Data Processing
- **Finance**: Brownian Motion, Black-Scholes, Arbitrage Pricing, Stochastic Calculus, Delta Hedging
- **Tools & Technologies**:
  - Dev: Git, GitHub/Lab, Docker, SQL, Linux, Bash, WSL2
  - Data Science/ML: PyTorch, NumPy, Pandas, NLTK, Dask, Scipy, Plotly, Matplotlib, Spacy, Scikit-learn, Seaborn, TensorBoard, AWS SageMaker, BeautifulSoup
  - Web: React, Flask, RESTful, Postman, Selenium
  - Other: ROS, LaTeX, Powershell, Memory Profiler
- **Soft Skills**: First-Principles Problem Solving, Leadership, Technical Writing, Self-teaching, Iterative Experimentation

Experience

Johns Hopkins University Applied Physics Laboratory

Computer Science Intern
Force Projection Sector: Ocean Systems & Engineering Group
May 2024 - Aug 2024 | Laurel, MD

- **Implemented** iteratively enhanced Generative Adversarial Imitation from Observation (GAIfO) agents, ***substantially outperforming*** baseline imitation models
- **Authored** critical literature reviews on GAIfO and Generative AI, providing ***direct insights*** for future project strategies.
- **Developed** an optimized GAIfO variant, using core-architectural insights from a literature review, which outperformed all prior versions over long timeframes
- **Enhanced** GTRI's SCRIMMAGE mass-simulation framework with increased complexity and expert controller functionality
- **Revamped** GitLab Continuous Integration pipelines, boosting speed and efficiency by ***25%*** while addressing security vulnerabilities
- **Optimized** project-wide Docker Image, used across all repositories, reducing pipeline build times by ***50%*** and increasing memory efficiency by ***40%***
- **Led** winning team for sector Intern Challenge in developing a secure, non-GPS intra-campus navigation prototype

University of Maryland MIND Lab

Research Intern, Breathing Analysis Project
October 2023 - Present | College Park, MD

- **Developed** an advanced visualization dashboard for efficient analysis of mass breath data
  - **Designed** dataset structures for visualization and feature extraction in future work
  - **Optimized** massive dataset-loading using Dask and multithreading by over ***400%***
- **Implemented** and evaluated supervised learning techniques for improved breath segmentation

University of Maryland CMNS

Student Researcher, Crowd Simulation
Advised by Dr. Sujeong Kim, Dr. David Mount
September 2024 - Present | College Park, MD

- **Exploring** application of non-Euclidean geometries
- **Applying** Transformers to crowd navigation,
  with focus on natural language goal-direction

Lead Teaching Assistant, CMSC351H (Algorithms Honors)
Spring 2024 | College Park, MD

- **Co-designed and graded** homeworks, exams, and lecture material for 38 honours students
- **Conducted** weekly office hours, providing personalized guidance on advanced topics

Publications

- *Ipelets for the Convex Polygonal Geometry*, published at SoCG 2024, LIPIcs Volume 293

Additional Qualifications

- **Certifications**: Complete Linear Algebra (Udemy), Algorithmic Toolbox (UCSD), Game Theory (Stanford)
- **Awards**: National Merit, Dean's Scholarship; Eagle Scout; Congressional App Challenge Winner; ISKF Black Belt
- **Languages**: English (Native), Bengali (Native), Hindi (Intermediate), Spanish (Intermediate), French (Beginner)